Exercise 5.24 Part 2

2. $\{12n + 9m : n, m \in \mathbb{Z}\} = \{3k : k \in \mathbb{Z}\}$.

Proof.

Denote $A = \{12n + 9m : n, m \in \mathbb{Z}\}$ and $B = \{3k : k \in \mathbb{Z}\}$.

($\subseteq$) By Fact **2.3**, we know for all integers x, y, then $4x + 3y$ is an integer. Now B has a subset $\{3(4x + 3y) : k = 4x + 3y\}$. Notice that such a subset is the same set as A. Thus, $A \subseteq B$.

($\supseteq$) We know that -1 times an integer is also an integer. Now A has a subset $\{12n - 9n : m = -n\}$. Notice that such a subset is the same set as B. Thus, $B \subseteq A$.

Now $A \subseteq B$ and $B \subseteq A$. Therefore, by Definition **5.19**, then $\{12n + 9m : n, m \in \mathbb{Z}\} = \{3k : k \in \mathbb{Z}\}$.

■

Exercise 5.24 Part 3

3. $\{5j + 1 : j \in \mathbb{Z}\} = \{5j + 6 : j \in \mathbb{Z}\}$.

Denote $A = \{5j + 1 : j \in \mathbb{Z}\}$ and $B = \{5j + 6 : j \in \mathbb{Z}\}$.

($\supseteq$) By Fact **2.3**, we know for all integers x, then $x + 1$ is an integer. Now A has a subset $\{5(x + 1) + 1 : j = x + 1\}$. Notice that such a subset is the same as B. Thus, $B \subseteq A$.

($\subseteq$) We know for all integers y, then $y - 1$ is an integer. Now B has a subset $\{5(y - 1) + 6 : j = y - 1\}$. Notice that such a subset is the same as A. Thus, $A \subseteq B$.

Now $A \subseteq B$ and $B \subseteq A$. Therefore, by Definition **5.19**, then $\{5j + 1 : j \in \mathbb{Z}\} = \{5j + 6 : j \in \mathbb{Z}\}$.

■